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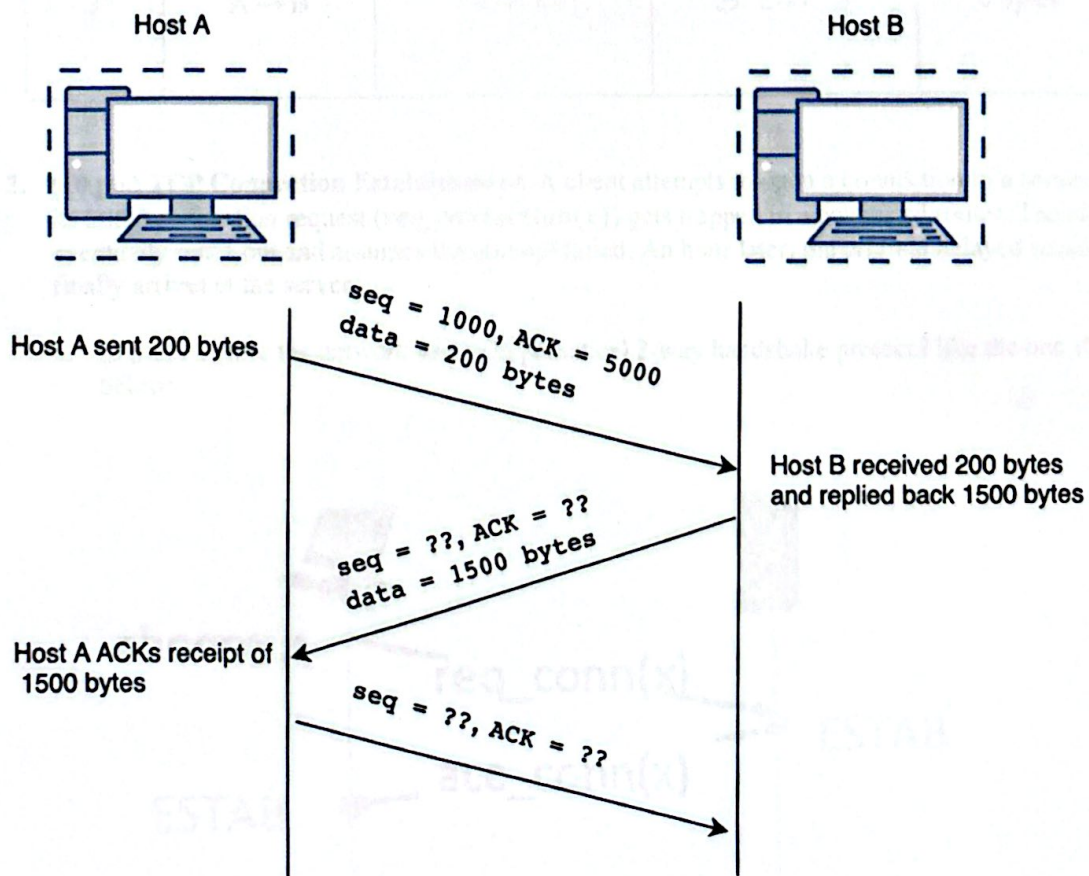
@ucsb.edu

Homework 02: DNS, CDN, Reliable Transport, TCP, UDP

Points: 100

- MAY ONLY BE TURNED IN ON GRADESCOPE as a PDF file.
 - There is NO MAKEUP for missed assignments.
 - We are strict about enforcing the LATE POLICY for all assignments (see syllabus).
- Only use the space provided for answers. Use clear and clean handwriting (or typing).
NOT USING THIS TEMPLATE WILL LOSE YOU 20 POINTS!**

1. (8 pts) **TCP Sequence and Acknowledgement Numbers.** The diagram below illustrates a TCP connection between Host A and Host B. The two hosts are exchanging data segments. Assume that no packets are lost and that the sequence numbers represent the byte stream as per standard TCP protocol rules.

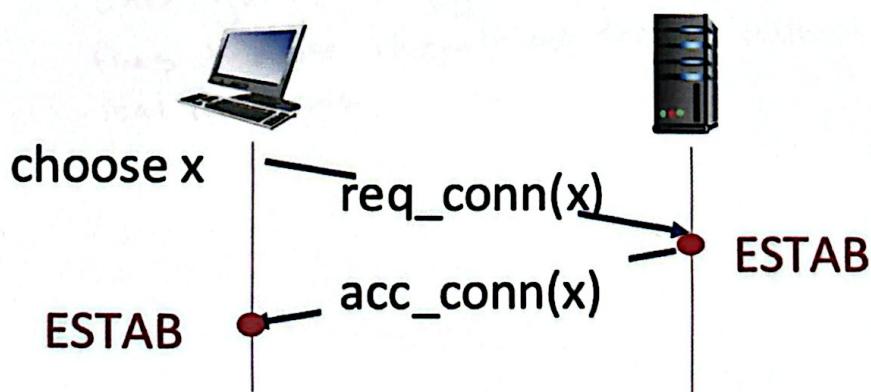


Using the information provided in the diagram, complete the missing fields in the table below. Calculate the exact **Sequence (Seq)** and **Acknowledgment (ACK)** numbers for the second and third segments in this exchange.

Segment	Direction	Sequence (Seq)	Acknowledgment (ACK)	Data Length
1	A → B	1000	5000	200 bytes
2	B → A	5000	1200	1500 bytes
3	A → B	1200	6500	0 bytes

2. (10 pts) **TCP Connection Establishment.** A client attempts to open a connection to a server, but its initial connection request (*req_connection(x)*) gets trapped in a congested router. The client eventually times out and assumes the attempt failed. An hour later, the original delayed request finally arrives at the server.

- a. (5 pts) Assume the network uses a hypothetical 2-way handshake protocol like the one shown below:



In this scenario, the server receives the delayed request and immediately transitions to the **ESTABLISHED (ESTAB)** state.

- i. (2 pts) Describe the impact this has on the server's limited resources: give two specific examples of what the server is now 'holding' for this non-existent client.

1. A receive buffer that allocates memory to buffer incoming data from the client
- this memory is unavailable for legitimate connections
2. A socket and corresponding TCP entry
<src IP, src port, dst IP, dst port> to use to send and receive packets from the abandoned client.

- ii. (3 pts) Explain why the server cannot simply 'reclaim' these resources immediately, even though the client has already timed out and disappeared.

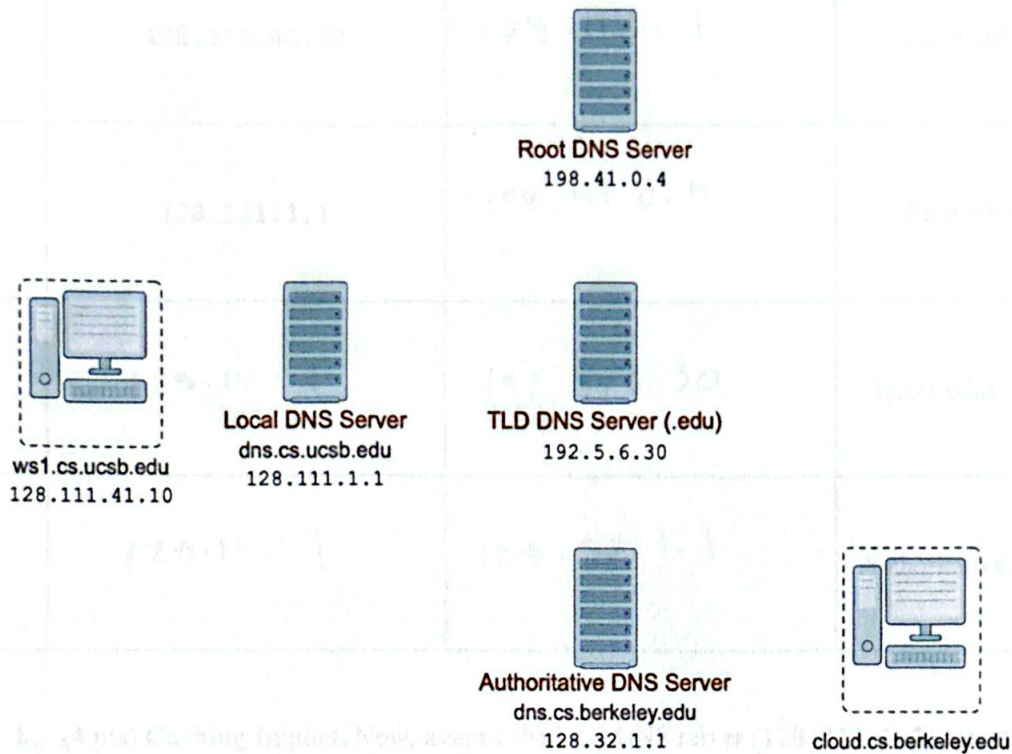
The server has no idea the client timed out. To the server, it looks like the client just hasn't sent over data yet. It simply has to wait until some idle timeout fires because it can't act sooner without risking breaking real connections.

- b. (5 pts) Standard TCP uses a 3-way handshake to solve this. Describe the single mechanical difference in the server's behavior that prevents the "waste" described in part (a). In your explanation, identify the specific state the server enters upon receiving the delayed request, the packet it sends back to the client, and the final condition that must be met before the server allocates full connection resources.

Instead of immediately starting to allocate resources, the server enters a "half-open" state where it is aware that a request came in but has not yet committed full resources to. It sends a message to the client and waits for an ACK in return to show that the client still wants the connection. This is the final condition that must be met before the server allocates full resources. The specific state is "wait for ack".

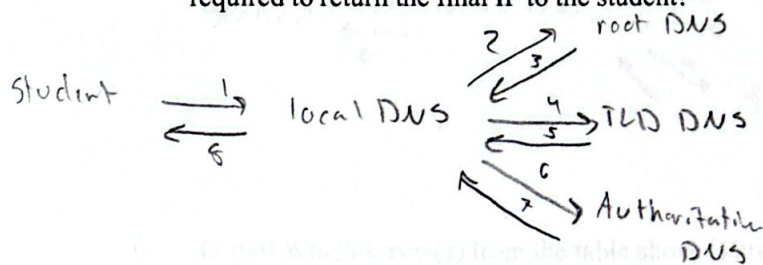
3. (15 pts) **DNS Resolution and Resource Records.** A student workstation at `cs.ucsb.edu` needs to resolve the IP address for `cloud.cs.berkeley.edu`.

- The workstation is configured to use its Local DNS Server (`dns.cs.ucsb.edu`) for recursive queries.
- The Local DNS Server performs all subsequent lookups via iterative queries.



a. (5 pts) **Iterative Resolution (No Caching).** Assume **no cached records** exist in the local DNS server except for the known root server IPs.

- i. (2 pts) How many **total DNS messages** (queries and responses combined) are required to return the final IP to the student?



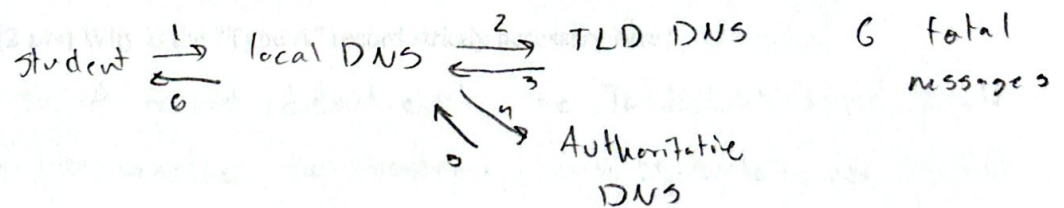
8 total DNS messages

- ii. (3 pts) Complete the table below for every DNS Request (query) message sent across the network.

Request #	Sender IP	Destination IP	Role of Destination Server
1	128.111.41.10	128.111.1.1	Local DNS Server
2	128.111.1.1	198.41.0.4	Root DNS Server
3	128.111.1.1	192.5.6.30	TLD (.edu) DNS Server
4	128.111.1.1	128.32.1.1	Authoritative DNS Server

- b. (4 pts) **Caching Impact.** Now, assume the local DNS server (128.111.1.1) recently resolved `www.berkeley.edu` and still has the `.edu` TLD DNS server record in its cache.

- i. (2 pts) How many **total messages** (queries and responses combined) are required now?



- ii. (2 pts) Which server(s) from the table above is/are **skipped** in this scenario?

The Root DNS server is skipped.

- c. (6 pts) **Resource Records (RR) and "Glue" Records.** In the part (b) scenario, the .edu TLD server receives the query but cannot provide the final IP of the target host. It must point the Local DNS server to Berkeley's authoritative server.

- i. (2 pts) What two specific types of DNS Resource Records (RR) must the .edu TLD server include in its response?

NS record - points to the authoritative DNS server's hostname

A record - gives the IP address of that nameserver

- ii. (2 pts) Provide the specific **Name** and **Value** fields for each record using the Berkeley information provided above.

NS record

- Name: cs.berkeley.edu
- Value: dns.cs.berkeley.edu

A record

- Name: dns.cs.berkeley.edu
- Value: 128.32.1.1

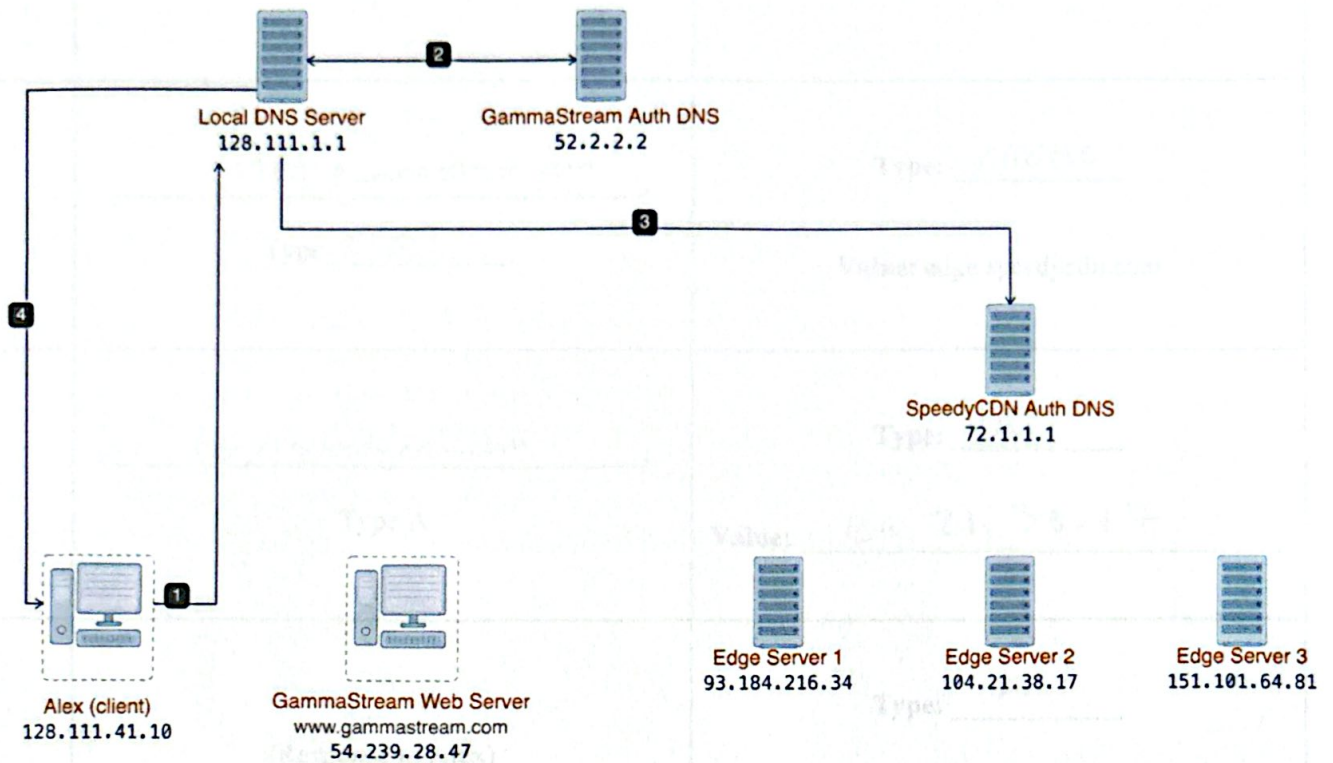
- iii. (2 pts) Why is the "Type A" record strictly necessary here?

If the A record didn't exist, the TLD DNS server would have to resolve the hostname dns.cs.berkeley.edu which requires asking for information from the same server that it was trying to gain information about in the first place, creating a circular dependency.

4. (12 pts) **CDN Redirection and Content Retrieval.** Alex (128.111.41.10) is visiting www.gamstream.com to watch a movie.

The page contains a link to the video: <http://video.gamstream.com/movie123.mp4>. GammaStream uses SpeedyCDN to serve its video content.

The network infrastructure is as shown below:



- a. (8 pts) Alex's host must resolve the IP address for the video content. Complete the following table for the DNS exchange. Assume Alex's Local DNS Server (128.111.1.1) already has the IP addresses for the Authoritative DNS servers of both [gamstream.com](http://www.gamstream.com) and speedycdn.com in its cache. It will contact these authoritative servers directly.

Note: When resolving the CDN hostname, SpeedyCDN's authoritative DNS server uses a **cluster selection strategy** (based on real-time network congestion and geography) and determines that **Server 2** is the optimal choice for Alex.

Interaction	Query (Name, Type)	Response (Type, Value/Answer)
1	video.gamstream.com, Type A	(Waiting for LDNS)
2	<u>video.gamstream.com</u> , Type <u>A</u>	Type: <u>cname</u> Value: edge.speedycln.com
3	<u>edge.speedycln.com</u> , Type A	Type: <u>A</u> Value: <u>104.21.38.17</u>
4	(Response to Alex)	Type: <u>A</u> Value: <u>104.21.38.17</u>

- b. (4 pts) After completing the DNS resolution, Alex's browser initiates a DASH-based stream. Fill in the blanks for the initial HTTP request message Alex sends to retrieve the video file.

Destination IP: 104.21.38.17

GET / movie123.mp4 HTTP/1.1

Host: video.gamstream.com

User-Agent: Mozilla/5.0

Accept: video/mp4

5. (18 pts) GBN vs. SR vs. TCP. Host A is sending a file to Host B.
- Each segment contains 1 byte of data.
 - Host A sends segments 1, 2, 3, 4, and 5 in order.
 - Segment 2 is lost in transit (as shown by the 'X' in the figure).
 - Assume the timeout interval is very long; Host A can send all segments and receive any generated ACKs before a timeout occurs.
 - Assume Host A's window is large enough to allow all 5 segments to be in flight.

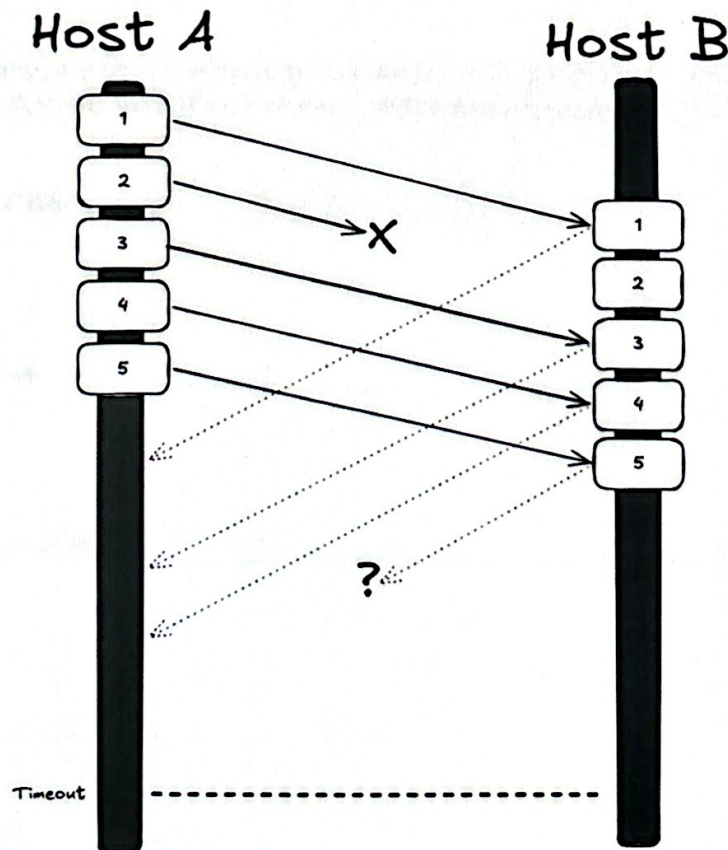


Figure: Segment 2 is Lost. ACK for Segment 5 will change based on the question part.

- a. (5 pts) For each protocol, identify the Acknowledgment (ACK) value sent by Host B in response to the arrival of segments 1, 3, and 4.

Arrival at Host B	GBN ACK Value	SR ACK Value	TCP Reno ACK Value
Segment 1 arrives	ACK 2	ACK 1	ACK 2
Segment 3 arrives	ACK 2	ACK 3	ACK 2
Segment 4 arrives	ACK 2	ACK 4	ACK 2

- b. (3 pts) Suppose Host A eventually receives the ACK generated by the arrival of Segment 5. According to the rules of each protocol, which specific segment(s) will Host A retransmit?

i. GBN: 2, 3, 4, 5

ii. SR: 2

iii. TCP Reno: 2

- c. (6 pts) Suppose a different scenario: Segment 1 is received, but its ACK is lost. However, Segment 2 is successfully received, and its ACK is received by Host A.

- i. (3 pts) Describe what happens to the sending window for GBN. Does it slide? To which segment?

The sliding window moves to segment 3 because if the sender receives Ack 2, it means segment 1 was already received since packets are sent and acknowledged in order.

- ii. (3 pts) Describe what happens to the sending window for SR. Does it slide?

No, the window stays as is since the window only moves when packet 1 is the smallest acknowledged packet. Since Ack 1 never made it all the way through, packet 1 is never properly acknowledged and the window cannot slide.

- d. (4 pts) Assume Host A is using TCP Reno.

- i. (2 pts) Compare Host A's behavior after it receives the ACK for Segment 4 versus when it receives the ACK for Segment 5. Identify any specific action(s) Host A takes at each point and explain the logic behind the timing.

TCP Reno's rule is to wait for 3 duplicate ACKs before acting.

The ACK for segment 4 will be the 2nd duplicate ACK, so the host won't act at this point.

The ACK for segment 5 will be the 3rd duplicate ACK, so Host A will retransmit segment 2 without waiting for a timeout.

- ii. (2 pts) What would happen to the missing Segment 2 if the ACK for Segment 5 was also lost in the network?

If the ACK for segment 5 gets lost in the network, then Host A never receives the 3rd duplicate ACK, so fast retransmit won't trigger. Host A would then have to wait for a time out before retransmitting segment 2.

6. (14 pts) **Multiplexing and Demultiplexing.** Three machines: **Host A**, **Host B**, and **Server S** are connected to the Internet.

- **Server S** is running two processes: a **UDP-based** chat application on **Port 5000** and a **TCP-based** web server on **Port 80**.
- **Host A** and **Host B** are both communicating with **Server S** simultaneously.

Network IP Addresses:

- **Host A:** 128.111.40.1
- **Host B:** 128.111.40.2
- **Server S:** 200.1.1.1

a. (6 pts) **Connectionless (UDP) Demultiplexing.** Suppose **Host A** and **Host B** both send a UDP message to the chat application on **Server S**.

- **Host A's** chat process uses **Source Port 3333**.
- **Host B's** chat process uses **Source Port 4444**.

i. (4 pts) Fill in the header information for the two UDP segments arriving at **Server S**.

can use the same port on server

Segment	Source IP	Src Port	Dest IP	Dest Port
From A	128.111.40.1	3333	200.1.1.1	5000
From B	128.111.40.2	4444	200.1.1.1	5000

ii. (2 pts) Does **Server S** use a separate socket to communicate with **Host A** and **Host B** for this UDP application? Explain using the concept of the **2-tuple**.

No, **Server S** does not use a separate socket to communicate with **Host A** and **Host B** because UDP allows the same socket on the server to send and receive messages from multiple hosts. The 2-tuple only has the destination IP & port since only one socket is needed to communicate with multiple sources. In this case the 2-tuple would contain the server's IP & port: (200.1.1.1, 5000).

b. (6 pts) **Connection-Oriented (TCP) Demultiplexing.** Now, Host A and Host B both open a TCP connection to the web server on Server S (Port 80).

- Host A's browser uses **Source Port 1111**.
- Host B's browser uses **Source Port 2222**.

i. (4 pts) Fill in the header information for the two TCP segments arriving at Server S.

Segment	Source IP	Src Port	Dest IP	Dest Port
From A	128.111.40.1	1111	200.1.1.1	80
From B	128.111.40.2	2222	200.1.1.1	80

ii. (2 pts) How many distinct sockets are currently active on Server S for Port 80? Explain your answer using the **4-tuple**.

There are 2 distinct sockets currently active on Server S because TCP requires different sockets be opened to communicate with different hosts. TCP uses a 4-tuple containing (src IP, src port, dst IP, dst port) to be able to identify each socket (src IP & src port are the keys since dst IP & dst port will be the same in both cases), allowing the server to maintain independent connections with each host.

- c. (2 pts) Complete the header information for the **response segment** sent from the TCP web server (Server S) back to **Host B**.

Segment	Source IP	Src Port	Dest IP	Dest Port
To B	200.1.1.1	80	128.111.40.2	2222

7. (8 pts) **True or False.** Indicate whether each statement is **True (T)** or **False (F)**.

- a. In the DASH protocol, the server is responsible for monitoring network congestion and automatically pushing a lower-quality video chunk to the client if bandwidth decreases.

False - client monitors bandwidth

- b. A major advantage of UDP over TCP is that it can begin transmitting application data without waiting for a round-trip time (RTT) to complete.

True - doesn't need handshake.

- c. If a receiver calculates a checksum that perfectly matches the value in the UDP checksum header field, it is mathematically guaranteed that the segment's data is uncorrupted.

False - it's possible there is a collision in checksums

- d. When a UDP receiver detects a checksum error, it sends a negative acknowledgment (NAK) back to the sender to request a retransmission.

False - discards the packet (NAKs only in TCP & rdt)

- e. The primary goal of TCP Flow Control is to prevent the network core (routers) from becoming congested with too many packets.

False - goal is to prevent the receiver's buffer from becoming overwhelmed

- f. Both TCP and UDP segment headers contain a field for the **Source Port Number**, even though UDP does not maintain connection state.

True - UDP - needed so receiver knows where to send a response
TCP - needed for demultiplexing

- g. To react as quickly as possible to packet loss, the TCP TimeoutInterval is set exactly equal to the most recently measured SampleRTT.

False - timeout is usually conservative and longer than an RTT. There is also variation in RTTs which could lead to unnecessary timeouts.

- h. A TCP receiver communicates its available buffer space to the sender by updating the rwnd (Receive Window) field in every acknowledgment it sends.

True - how TCP flow control works. sender limits how much data in flight to stay within the value.

(15 pts) Wireshark Lab Handin

Please follow the Wireshark lab writeup and provide your answers to the following questions.

Part 2.1:

1. (1 pt) Locate the DNS query and response messages. Are they sent over UDP or TCP?

UDP

2. (2 pts) What is the destination port for the DNS query message? What is the source port of DNS response message?

query 1 : src: 54335 dst: 53
 query 2 : src: 57975 dst: 53
 query 3 : src: 45349 dst: 53
 response 1 : src: 53 dst: 45349
 response 2 : src: 53 dst: 54335
 response 3 : src: 53 dst: 57975

3. (2 pts) To what IP address is the DNS query message sent? This is the IP address of a local DNS server.

128.111.1.1

4. (2 pts) Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?

Type A. The query message does not contain any answers

Part 2.2:

1. (2 pts) What is the destination port for the DNS query message? What is the source port of the DNS response message?

query: src: 52896 dst: 53
response: src: 53 dst: 52896

2. (1 pt) To what IP address is the DNS query message sent?

128.111.1.2

3. (2 pts) Examine the DNS response message. How many "answers" are provided? What do each of these answers contain?

3 answers

1. Type cname. contains the canonical name for www.mit.edu
2. Type cname. contains the canonical name for the canonical name of www.mit.net
3. Type A. contains the IP address of the cname from answer 2.

Part 2.3:

1. (3 pts) Examine the DNS response message. What MIT nameservers does the response message provide? Does this response message also provide the IP addresses of the MIT nameservers?

Provides 8 MIT name servers but does not provide the IP addresses of said name servers

1. asia1.akam.net
2. eur5.akam.net
3. ns1-37.akam.net
4. asia2.akam.net
5. use2.akam.net
6. use5.akam.net
7. usw2.akam.net
8. ns1-173.akam.net